

# Draft Report: Predicting Momentum Shifts in Competitive E-sports

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**Course:** DS680

**Topic:** Temporal Sentiment Analysis of HLTV Community Discourse

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## 1. Abstract (Completed)

This research investigates the predictive power of community sentiment "velocity" within the professional e-sports ecosystem. By analyzing live-scraped data from HLTV.org match threads during professional Counter-Strike 2 (CS2) matches, this study compares the performance of a baseline N-gram Naive Bayes model and a Deep Learning LSTM architecture. Preliminary results suggest that rapid shifts in sentiment polarity—defined as "Sentiment Velocity"—serve as a leading indicator for in-game momentum shifts, such as comebacks and "chokes," prior to scoreline updates.

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## 2. Introduction (Completed)

Sentiment analysis has traditionally been applied to static datasets, such as movie reviews, where the objective is to categorize a fixed opinion. However, in the high-velocity environment of competitive e-sports, sentiment is fluid and reactive. This project explores the "Wisdom of the Crowd" by quantifying how the collective emotional state of thousands of spectators correlates with match outcomes. The core hypothesis is that a significant increase in negative sentiment velocity can predict a "momentum collapse" for a leading team, while positive velocity predicts a successful "clutch" or comeback.

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## 3. Related Work (Literature Review) (Completed)

Based on current NLP research, this study identifies three key pillars:

- **Momentum Forecasting:** Research by *WSDM* indicates that psychological momentum in e-sports can be modeled as a non-linear fluctuation detectable by Recurrent Neural Networks (RNNs).
- **Audience Engagement:** Recent studies highlight that real-time chat engagement is highly sensitive to in-game "turning points," such as objective completions or pivotal kills.

- **Wisdom of the Crowd:** Evidence suggests that crowd sentiment derived from social platforms can outperform professional projections by nearly 30% in sports analytics contexts.

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## 4. Methodology (Completed)

### 4.1 Dataset and Preprocessing

The dataset consists of timestamped comments scraped from **HLTV.org**. To ensure data integrity, the following nuances were implemented:

- **Regex Sanitation:** Custom Regular Expressions were used to strip `<br />` tags and gaming-specific emoticons.
- **Normalization:** All text was converted to lowercase to consolidate the vocabulary.
- **Reproducibility:** An **80/20 train-test split** was applied with a `random_state` of 42.

### 4.2 Model Architectures

1. **Baseline (N-gram Naive Bayes):** A probabilistic model using a Bag-of-Words approach with **N-grams (1,2)** to capture local context like negation.
  2. **Advanced (LSTM):** A sequential model featuring an **Embedding Layer** and **64 LSTM units** to track sentiment across the timeline of a match.
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## 5. Preliminary Results and Analysis (In Progress)

- **Baseline Performance:** Initial tests on the N-gram Naive Bayes model show an accuracy of approximately **86.84%** in identifying general sentiment.
- **Deep Learning Gains:** The LSTM model currently demonstrates an accuracy of **88.13%**, proving superior at capturing the sequential flow of forum "hype."
- **Velocity Visualization:** (Preliminary Graph) Matches with high "Negative Velocity" ( $V = \frac{\Delta \text{Sentiment}}{\Delta \text{Time}}$ ) frequently preceded a team losing more than 4 consecutive rounds.

[Placeholder: Insert HLTV Sentiment Velocity vs. Round Score Chart]

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## 6. Discussion and Future Work (Outline)

- **Strengths:** The LSTM effectively handles word order and long-term dependencies.
- **Weaknesses:** Both models currently struggle with "Confident Errors" where irony or "copy-pasta" slang is misinterpreted as genuine sentiment.
- **Future Work:** Implementing Transformer models (BERT) and Early Stopping to reduce overfitting.

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## 7. References

- *Scalable Psychological Momentum Forecasting in Esports*, WSDM (2020).
- *Analyzing Audience Engagement in Esports*, ResearchGate (2025).
- *Sports Analytics with NLP: Using Crowd Sentiment to Help Pick Winners*, Harvard (2022).

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### Action Plan for Final Submission:

1. **Finalize Analysis:** Complete the correlation study between Velocity scores and HLTV match logs.
2. **Visualization:** Generate the final "Velocity vs. Momentum" plots to include in the Results section.
3. **Refine Related Work:** Expand the critique of the 3-5 research papers to meet the page requirement.